Passenger Vehicle Supply & Demand Analysis: Final Report

Time Series Analysis

07 December 2025

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Summary

This report leverages multiple economic time series data available on the Federal Reserve Bank of St. Louis (FRED) web server to develop a mechanism for understanding vehicular supply and demand within the United States. It analyses economic, political, and manufacturing decisions by using two separate frameworks.

Framework 1 conducts an in-depth analysis of Vehicle Miles Traveled (VMT), Manufacturer's New Orders (AMVPNO) and public transportation usage (TRANSIT) as the primary response variables. Four separate modeling techniques, including General Additive Models (GAMs), Seasonal Autoregressive Integrated Moving Average with Exogenous Factors (SARIMAX), Vector Autoregressive Modeling with Exogenous Factors (VARX) and Extreme Gradient Boosting (XGBoost) were incorporated into the analysis to determine the optimal modeling technique. Root Mean Squared Error (RMSE) and Mean Absolute Percentage Error (MAPE) were used as the primary evaluation criteria across models, with Akaike and Bayesian Information Criterion (AIC/BIC) used within models to determine the best competitor. XGBoost consistently outperformed all other models in the prediction of VMT, AMVPNO, and TRANSIT. Due to the "black box" nature of XGBoost, a supplementary analysis employed a SARIMAX - General Autoregressive Conditional Heteroskedasticity (GARCH) module, independent of previous work. It examined VMT dynamics and their sensitivity to macroeconomic indicators. VIF analysis was used to confirm the stability of the selected exogenous predictors. The SARIMAX-GARCH(1,1) captured time-varying volatility and reproduced recent VMT trends. It also simulated economic shocks of 20% to evaluate its robustness to world events and economic crashes. However, due to XGBoost's superior performance, it was further implemented in Framework 2.

Framework 2 applied an independent XGBoost model to forecast Total Vehicle Sales (TOTALSA), leveraging the algorithm's capacity to capture nonlinear relationships and complex lag structures within economic time series. To capture temporal patterns, the model used autoregressive inputs which included 1, 2, 3, 6 and 12 month lags of TOTALSA as well as 1, 3 and 6 month lags for the other exogenous variables mentioned in Table 1 . Using time-series cross-validation, grid-based hyperparameter tuning, and early stopping, the model achieved strong out-of-sample accuracy over a two year test period, with a RMSE of 0.54 and a MAPE of 2.67%. Feature importance results revealed that TOTALSA is predominantly autoregressive, with the one month lag contributing approximately 77% of the model's predictive gain, while exogenous variables provided smaller but meaningful contributions. Overall, Framework 2 demonstrates that XGBoost is highly effective for modeling complex, nonlinear sales dynamics and offers strong short-term forecasting performance despite limited interpretability.

Introduction

Transportation remains at the heart of the American way of life. The U.S. Census Bureau (USCB) estimates that the average commute to work is approximately 27.2 minutes, one-way, in the United States (2025), with approximately 9.3% of Americans traveling greater than 60 minutes, one-way, to get to their place of work (USCB, 2025). As the population increases and suburban sprawl spreads across the United States, researchers consistently find that neighborhoods are increasingly untraversable without a vehicle (King & Clarke, 2015). Considering 69.2% of Americans travel alone to work (USCB, 2025), it is increasingly valuable to examine the accessibility of vehicles to the public. Thus, this analysis serves as a retrospective review of vehicle supply and consumer demand between January of 2000 and 2024.

Previous research on vehicle usage and consumption is focused on factor analysis using base regression techniques. Demographic and socioeconomic factors directly correlate to the newer generation's requirement to own a vehicle. An example is suburban sprawl; due to the exorbitant cost of city living, affordable housing, and the remoteness of suburban housing, access to passenger vehicles has become a staple to maintain employment since the early 2000s (Resnik, 2010). Access to online platforms, such as Carvana, also shapes how the newer generations interact and shop for private automobiles. For example, the purchasing habits of Generation Z show that online brokers are the preferred method of shopping for vehicles due to its accessibility and straightforward pricing (Whitaker et al., 2024). Additionally, priorities in car choice shift with generation - rather than focus on fuel and purchase cost, Nerurkar et al. (2024) suggests that current consumers are more fashion and environmentally focused.

These demand factors can also be mitigated by other convoluting variables. Keith et al (2024) demonstrated that even though utilization requirements increased, the development of ride-share platforms and mobile applications reduced the need to purchase an individual motor vehicle. Equivalently, improved infrastructure for public transportation or bike lanes can also greatly reduce the need to own or purchase a personal vehicle (Younes, et al., 2014).

In this regard, modeling of vehicle usage as a time series is a novel concept. Several quantifiable aspects about vehicle usage, such as miles traveled, vehicles sold, and units produced have natural analogs to trend and seasonality. Modeling as a time series omits the necessity to overfit for all potential confounding variables that are not easily captured in datasets. While rarely used, the efficacy of this methodology was demonstrated in Chen (2024). Thus, time series analysis will be used to forecast vehicle usage, determine trends between supply and demand, and identify potential exogenous factors influencing vehicle sales. While existing studies have analyzed consumer preferences, production, or policy factors in isolation, few have integrated these dimensions into a unified forecasting framework. This project will look to address that gap by combining macroeconomic, behavioral, and policy data to model the dynamic interactions of vehicle supply and demand over time.

Data Summary

The datasets used in this analysis were collected through the Federal Reserve Bank of St. Louis (FRED). The primary source varies for each series and are provided in the References section. A general overview of the data considered in this analysis is provided below in Table 1. All data considered was evaluated on a monthly frequency. Although observations dating to the early 1960s exist, only the timeframe between 2000-2024 was considered:

Table 1: An overview of the potential variables analyzed throughout this paper. All variables contained complete (0 NA) observations for the monthly periods of Jan 2000-2024.

Series	Description	Units
TOTALSA	Total sales in the US	Millions of units
AMVPNO	Manufacturers' new orders	Millions of USD
FAUTOSA	Foreign auto retail sales	Thousands of units
DAUTOSAAR	Domestic vehicle sales	Millions of units
DAUPSA	Domestic auto production in USA	Thousands of units
VMT	Vehicle miles traveled	Millions of miles
AUENSA	Auto exports (from US)	Thousands of units
NATURALGASD11	Natural gas consumption	Billion Cubic Feet
TRANSIT	Trips on public transportation	Thousands of unlinked trips
GASREGCOVM	Conventional gas price	Dollars/Gallon

AISRSA	Inventory: Sales ratio	%

To visualize the comparative variability of the different series, all series were normalized using the Z-score methodology. The visualization is provided below in Figure 1.

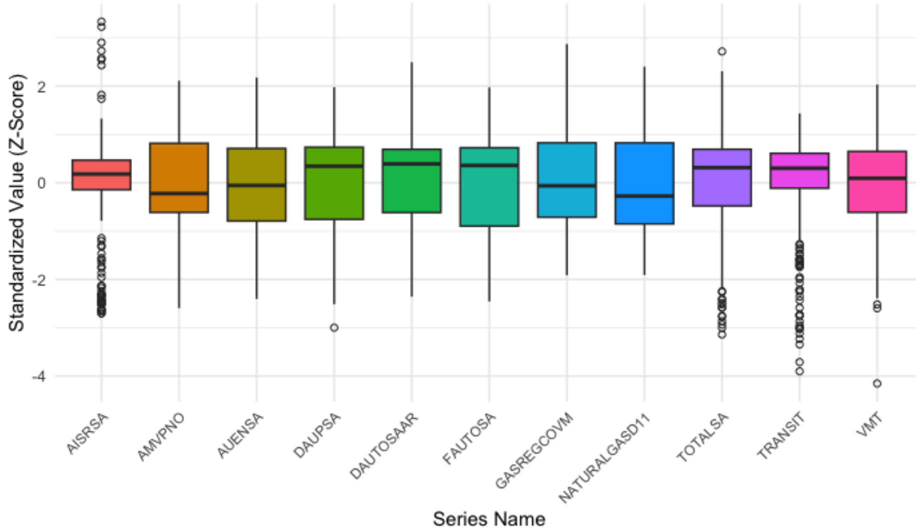

Figure 1: Z-Standardized Boxplots of all variables examined.

The raw time series plots (Figure 12 and Figure 13, in Graphics) show large volatility around the 2008 U.S. financial crisis and the COVID-19 pandemic. This accounts for the significant outliers seen in the AISRSA, TOTALSA, and TRANSIT series. Variables like DAUPSA, DAUTOSAAR, FAUTOSA, and TOTALSA are noticeably skewed. The Autocorrelation Function (ACF) plots (Figure 14, in Graphics) show strong correlation extending well beyond the confidence bands. To formally determine if the data needs transforming and differencing, Table 2 summarizes the results from the Box-Cox transformation and the Augmented Dickey-Fuller (ADF) tests.

Table 2: Summary of Box-Cox lambda and ADF p-values for all series.

Series	**Box-Cox (λ)**	**Transformation**	**ADF (p-val)**
TOTALSA	1.999	y^{λ}	0.510
AMVPNO	-0.054	Log(y)	0.451
FAUTOSA	0.571	$\sqrt{y}$	0.355
DAUTOSAAR	1.062	N/A	0.701
DAUPSA	1.807	y^{λ}	0.446
VMT	0.759	N/A	0.01
NATURALGASD11	0.779	N/A	0.137
TRANSIT	1.999	y^{λ}	0.461
GASREGCOVM	-0.498	$y^{-\lambda}$	0.442
AISRSA	0.578	$\sqrt{y}$	0.198

The Box-Cox transformation and a first order differencing ($\Delta\alpha_t = \alpha_t - \alpha_{t-1}$) were applied to the examined series. From this, a true correlation matrix for the variable interplay (Figure 2) was developed; due to the stated transformations, a high correlation here implies a true linear relationship for rate of change. This provides a valuable framework for examining exogenous variables; for example, we can determine that public transportation (TRANSIT) has the strongest, positive correlation with vehicle miles traveled (VMT).

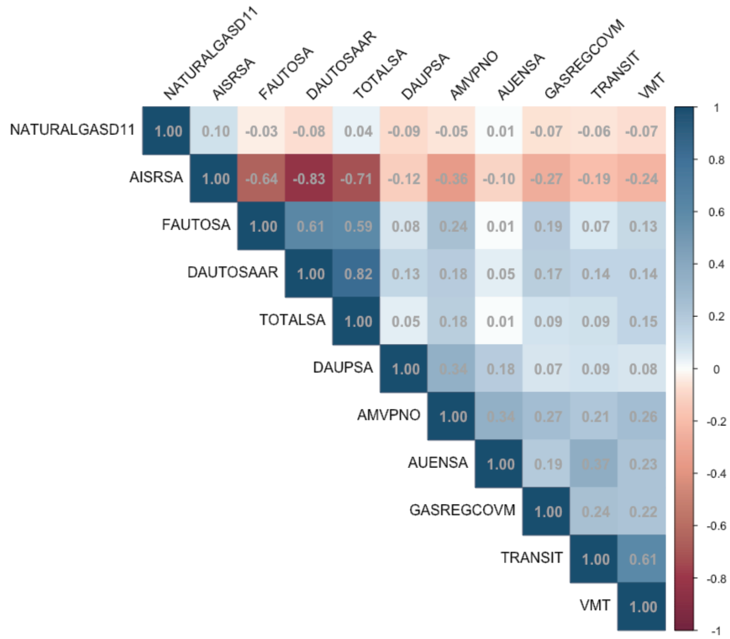

Figure 2: Correlation matrix of transformed and differenced data.

Analysis

Using the initial diagnostic steps above, we continued the analysis using automated model fitting methods in the R statistical software. The analysis focused on using separate models with VMT, AMVPNO, and TRANSIT used as response variables to determine variable relationships. The three response variables were deliberately selected to represent distinct but interconnected facets of the passenger vehicle ecosystem:

Vehicle Miles Traveled (VMT) serves as the most direct proxy for actual consumer utilization and demand for personal vehicle usage. It captures real-world driving behavior and is sensitive to macroeconomic conditions, fuel prices, remote work trends, and substitution with public transit.

Manufacturers' New Orders: Motor Vehicles and Parts (AMVPNO) is widely regarded as a leading indicator of automotive supply response. Unlike production or retail sales (which can be satisfied from existing inventory), new orders reflect manufacturers' forward-looking decisions and are therefore the earliest signal in the supply chain adjustment process.

Public Transit Ridership (TRANSIT) was included as the primary substitute good. Economic theory and empirical evidence suggest an inverse relationship with private vehicle usage: when public transit is attractive (or private driving becomes costly), consumers shift modes, reducing both VMT and the long-run need for new vehicles.

Methods

Four approaches were considered in the data analysis process. First, we used General Additive Models. This stems from the concept that a time series can be summarized as an additive model consisting of a smooth, non-linear trend, seasonal or periodic variation, and random variation.

$$Y_t = s(x_t) + \sum_m I_m * \beta_m + \epsilon_t$$

$$where\ I_m = \{1,\ \ if\ x_i\ occurs\ during\ month\ m, 0\ otherwise$$

In this modeling approach, we examined the time series as univariate phenomenon. A fixed, monthly variable was used for each examined response variable. The smoothing function $s(x_i)$ models a non-linear trend within the time series. As GAMs are ultimately linear models, they do still rely on the independence of observations, which creates issue with the temporal nature of a time series. This would be mitigated through an autoregressive, or recursive, term. The fixed, monthly variable indicator models for the seasonal effect.

We address the serial correlation using the Seasonal Autoregressive Integrated Moving Average with Exogenous Factors (SARIMAX) methodology.

$$SARIMAX(p, d, q)x(P, D, Q)_m$$

SARIMA incorporates both an ARIMA (p, d, q), non-seasonal component, and a seasonal (P, D, Q) component. The parameters are: p, the order of autoregressive terms; d, the order of differencing required to achieve stationarity; and q, the number of past error terms (moving average) to improve the forecasted value. The uppercase parameters are unique to the fixed period seasonality (m = monthly). Finally, the Exogenous Factors consist of a linear regression of selected external variables applied to the dependent variable, alongside the SARIMA terms. As a general linear model, SARIMAX requires the time series to be stationary, which is accomplished through the integrated, or differencing (d, D), parameters. Furthermore, it also requires the residuals to be independently distributed, which was formally evaluated using inference testing like the Ljung-Box test.

Next, the Vector Autoregressive Exogenous (VARX) model was applied to the three response variables of interest. VARX provides a multivariate time series analysis model that allows external factors to improve forecasting.

$$Y_t = C + \sum_i^p (\phi_i Y_{t-i}) + \sum_{j=0}^r B_j X_{t-j} + \epsilon_t$$

$$\begin{aligned} where{:}\ Y_t &= Endogenous\ Vector\ (k\ variables), X_t = Exogenous\ Vector\ (m\ variables), \epsilon_t \\ &= Residual\ Vector\ (k\ endogenous\ terms) \end{aligned}$$

This multivariate model enables simultaneous forecasting for all variables and shows how the variables interact with each other. For the model to function, all examined time series ($\mathbf{Y}_t$ and $\mathbf{X}_t$) must be stationary prior to function. The residuals must also resemble white noise. Finally, the errors across the different equations (as this is vectorized) may exhibit contemporaneous correlation.

The final model applied was Extreme Gradient Boosting (XGBoost). XGBoost is a non-parametric, non-linear, machine learning algorithm that provides efficient outputs for both regression and classification problems. It recursively uses weak learners, typically decision trees, to minimize the error terms of the previous weak learner. Then, it uses an amalgamation, or ensemble, of learners until the overall stopping criterion is met. While not designed for temporal datasets, artificial features were created to mimic the time component of the series (month, year, et cetera). Encoding temporal features into a dataset has been an effective way to implement XGBoost, as seen in Shobayo et al. (2025) and Dama & Sinoquet (2021). The primary limitation of XGBoost is its non-parametric and non-linear methodology; this generates a black box in decision making, granting very minimal insights into the relationships between the variables. The variable interactions can marginally be understood by using the reported feature importance output, which due to the feature imputation is still difficult to interpret.

In addition to the XGBoost models developed for VMT, AMVPNO and TRANSIT a separate XGBoost modeling framework was constructed to forecast Total Vehicle Sales (TOTALSA). Since, this variable was not included as one of the primary response variables in the previous model an independent workflow was required. XGBoost was selected due to its superior performance compared to the other models in the previous framework as well as its demonstrated ability to capture non-linear relationships, handle complex feature interactions and perform well in high-dimensional structured data environments (Xia et al., 2020).

This approach incorporated an expanded feature-engineering strategy designed to emulate the autoregressive behavior present in traditional time-series models. Lagged versions of the Total Sales series (1, 2, 3, 6, and 12 month lags) were created, along with short, medium and long term lag values (1, 3 and 6 month lags) of all the other exogenous indicators which included production, exports, demand, transit usage, and fuel costs. Additionally, calendar based variables, including month, quarter, and a year-over-year sales growth feature, were added to account for seasonality and cyclical purchasing behavior. To ensure robust prediction, the dataset was split into a training and a 24-month test set, preserving temporal order. A hyperparameter tuning process which used grid search, time-series cross-validation, and early stopping was implemented to identify an optimal model configuration. This allowed the Total Sales XGBoost model to incorporate both short-term autocorrelation and complex non-linear relationships among macroeconomic factors. The best configuration was selected based on minimum test RMSE and early stopping to prevent overfitting. Final model training was then completed using these obtained optimized parameters.

Results

Framework 1

Table 3: Summary of methods and outputs for different methodologies.

Main Response	Relationship Modeled	Method	Root Mean Squared Error (RMSE)	Mean Absolute Percentage Error (MAPE)	Fitting Metric
VMT	VMT ~ s(pts) + month	GAM	19411.49	6.39	R2 = 0.87
	VMT ~ ARIMA(1,0,3)(0,1,1)[12] + XReg: (TRANSIT)	SARIMAX	494.96	2.48	AIC = 3621.33 BIC = 3649.59 LB pval = 0.11
	VMT	XGBOOST	291.31	1.45	N/A
AMVPNO	AMVPNO ~ s(pts) + month	GAM	9302.89	14.17	R2 = 0.88
	AMVPNO ~ ARIMA(0,1,3) + XREG: (AISRSA)	SARIMAX	0.10	1.08	AIC = -943.67 BIC = -925.79 LB pval = 0.82
	AMVPNO	XGBOOST	0.08	0.99	N/A
TRANSIT	TRANSIT ~ s(pts) + month	GAM	177879.4	28.56	R2 = 0.93
	Log(TRANSIT) ~ ARIMA(1,1,2)(1,0,0)[12]+ Xreg: (AUENSA)	SARIMAX	0.46	1.68	AIC = -183.5 BIC = -162.05 LB pval = 0.01
	Log(TRANSIT) ~ ARIMA(2,0,1)(1,0,0)[12] + XReg: (AUENSA, COVID)	SARIMAX	0.73	2.60	AIC = -288.8 BIC = -260.16 LB pval = 0.45
	TRANSIT	XGBOOST	0.36	1.26	N/A
Endogenous	VMT	VARX	400.68	2.05	AIC = 1.33 BIC = 2.19 LB pval = 0.01
	AMVPNO	VARX	0.25	2.67	
	Log(TRANSIT)	VARX	0.48	1.60	

Table 3 provides a high-level summary of all examined models for the three target variables: VMT, AMVPNO, and TRANSIT. A consistent trend was observed in model complexity versus interpretability. The overall predictive accuracy of the models, measured through test Root Mean Squared Error (RMSE) and Mean Absolute Percentage Error (MAPE), were consistently in the order of:

$$GAM < SARIMAX \approx VARX < XGBoost$$

The series of General Additive Models served the intended purpose of providing a baseline level of comparison for the more complex models. As predicted, the GAMs' inability to account for serial correlation resulted in a large magnitude of bias. The incorporation of autoregressive features, as well as exogenous variables, significantly improved forecasting abilities of the models. From the results in Table 3, we gained the following insights: Vehicle Miles Traveled (VMT) was successfully modeled by its most correlated variable, public transportation trips (TRANSIT). Even with the weakest model, GAM, the MAPE for VMT reached a high of only 6.39%, decreasing to 1.45% as the models became increasingly complex. The pictorial GAM for VMT reveals why GAM generally performs poorly for all response variables; it is a global under estimator. Additionally, it is completely incapable of reacting to exogenous shocks, most evident during the pandemic. Referring to Figure 3; the model fit during the COVID-19 pandemic does not react fast enough to model the dramatic reduction in vehicle miles traveled. On the contrary, SARIMAX, VARX and XGBoost all closely follow the two-year test period. All models exhibit an almost 90% reduction in RMSE compared to GAM. SARIMAX effectively modeled American VMT and met the model assumptions of white noise residuals (p = 0.11).

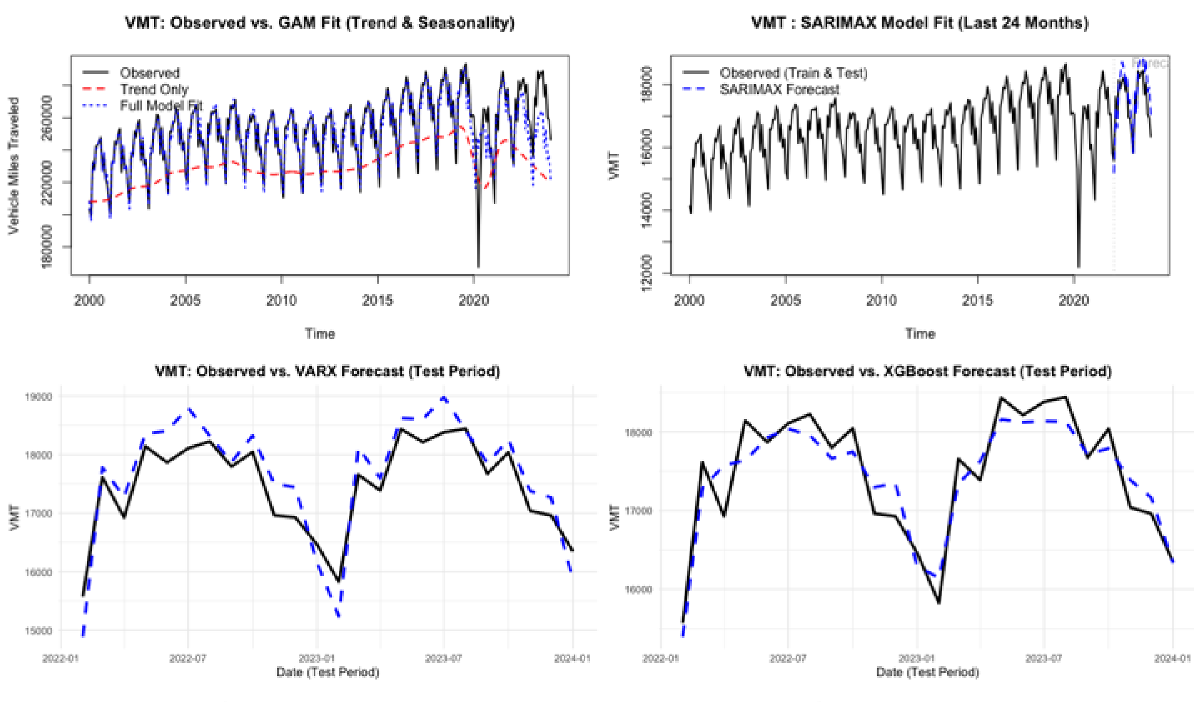

Figure 3: Comparison of model predictive performance for VMT models.

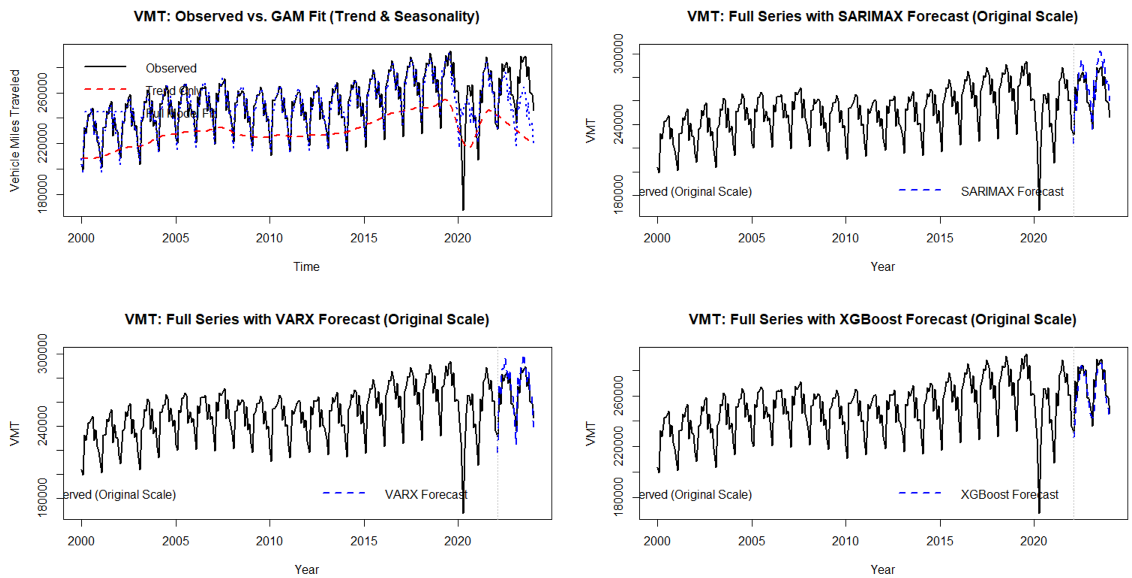

Figure 4: Comparison of model predictive performance for VMT models (original scale).

Manufacturer's New Orders (AMVPNO) primary exogenous driver, US Inventory to Sales Ratio (AISRSA), yielded a MAPE of 1.08% and a high p-value (0.82) from its Ljung-Box test. This provides a direct connection to supply and demand relationships in American automotive production. Referring to Figure 5; The GAM for AMVPNO faces the same issues as it did with VMT; the model reacts too slowly to exogenous changes, showing a great degree of bias during the COVID-19 pandemic. Notably, the units (millions) used for AMVPNO obfuscate the comparative performance of the remaining model. SARIMAX displays a great degree of variability, VARX is a global over estimator, and XGBoost is a global under estimator. Due to the relatively low difference in MAPE, and that SARIMAX and VARX have uncorrected trends over the course of two years, XGBoost is the most reliable model for forecasting.

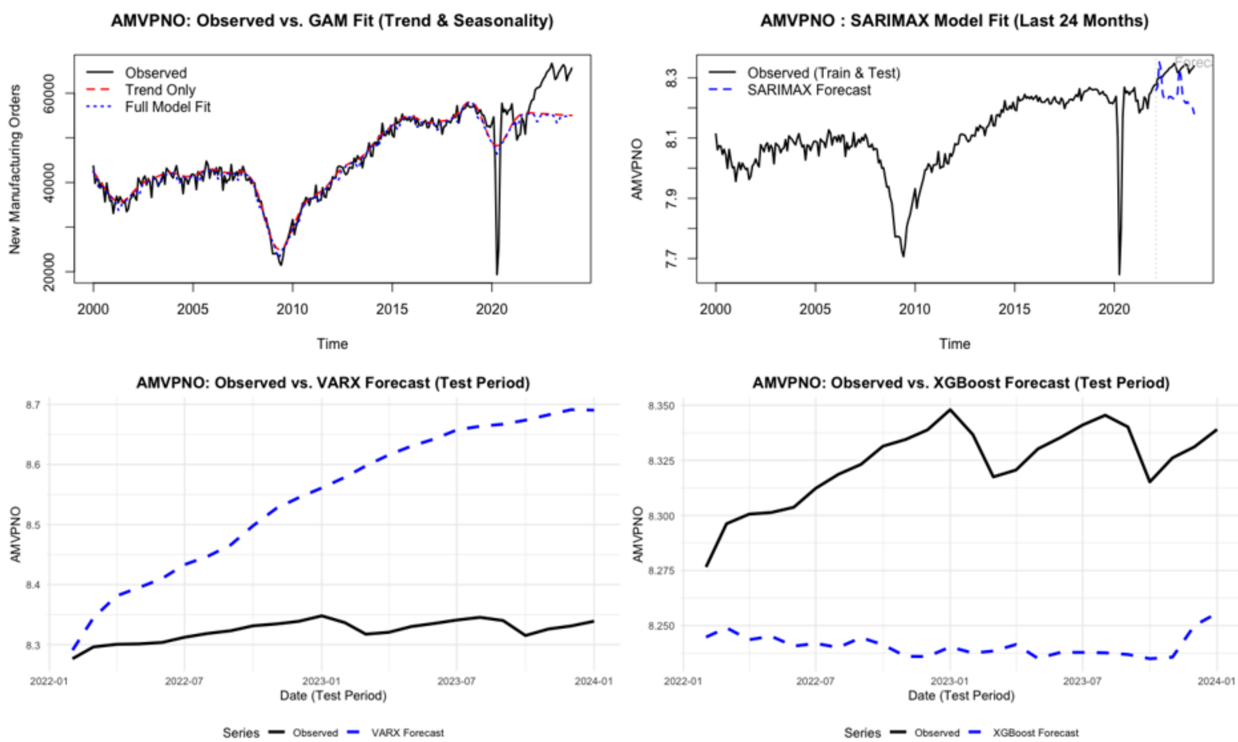

Figure 5: Comparison of model predictive performance for AMVPNO models.

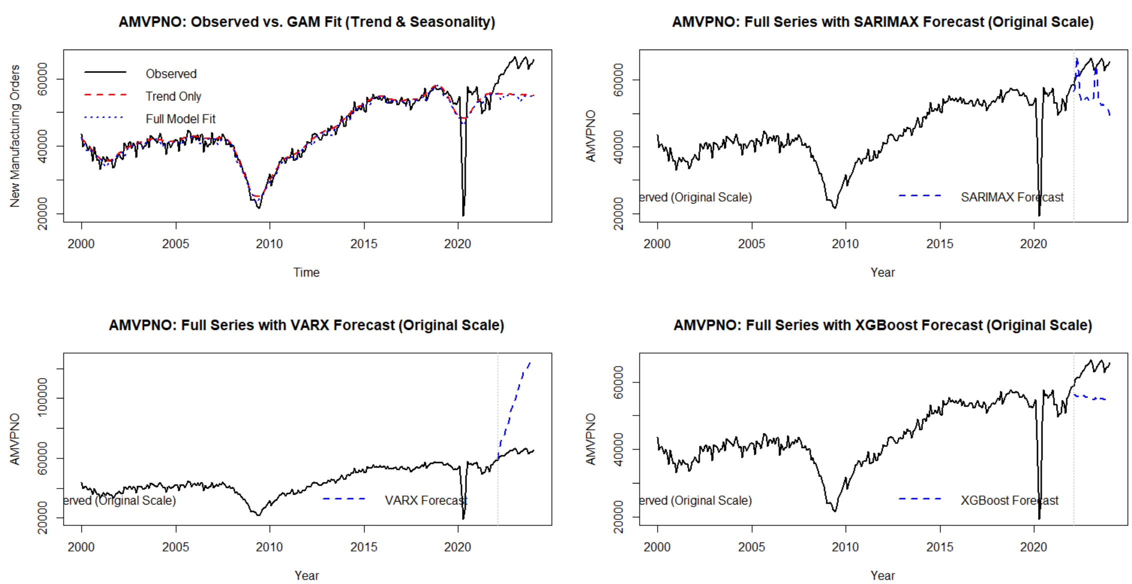

Figure 6: Comparison of model predictive performance for AMVPNO models (original scale).

TRANSIT posed the greatest modeling challenge. Despite the initial Box-Cox transformations, TRANSIT models continuously failed the Ljung-Box test (p-value ≤ 0.01). Attempted corrections included: Box-Cox transformation of the TRANSIT variable, mandated differencing in the ARIMA fit, additional transformation in the TRANSIT variable, and finally the inclusion of the COVID-19 dummy variable and AUENSA as an exogenous variable. One of these failed models is included as an example in Table 3. The COVID-19 variable was set to 1 for all days the United States was officially in a national emergency due to the pandemic, and 0 otherwise.

Once these two changes were incorporated, white noise residuals were observed (p = 0.45). Figure 7 provides a visual comparison of forecasting for all TRANSIT models.

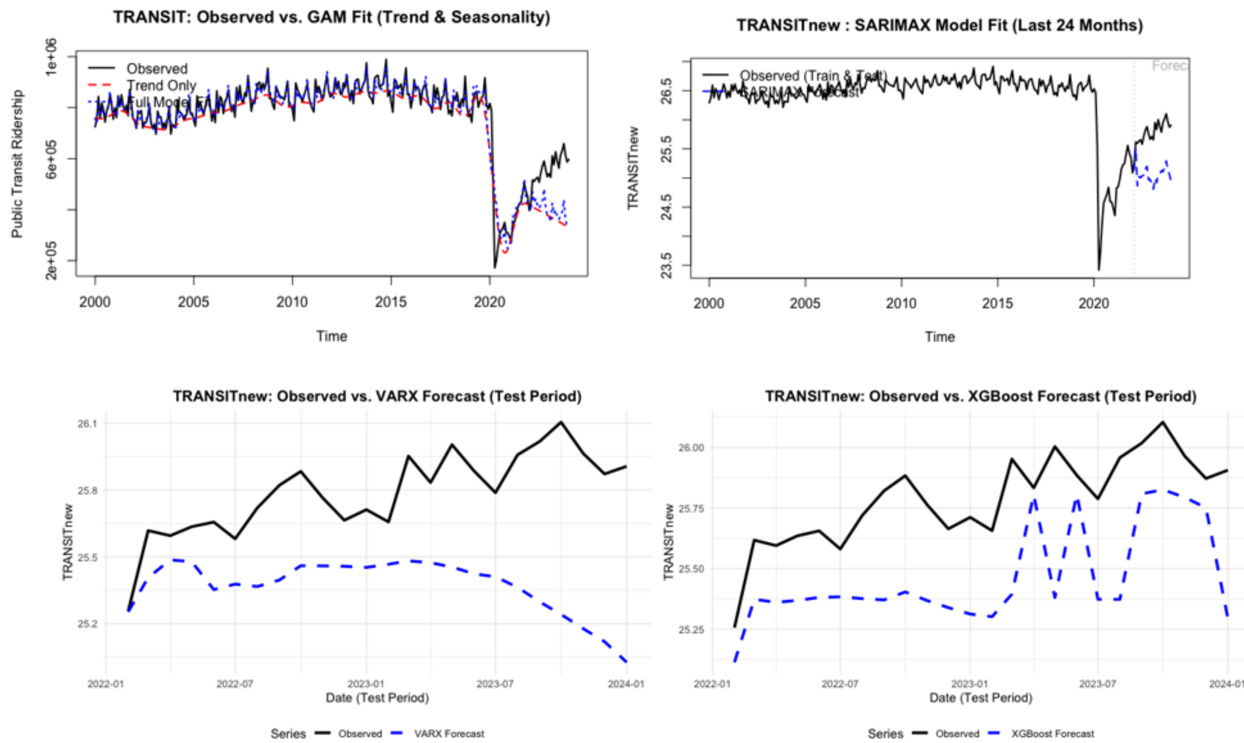

Figure 7: Comparison of model predictive performance for TRANSIT models.

Due to the logarithmic transformation of the TRANSIT variable, the range of the response variable is greatly shortened. This, like with AMVPNO, obfuscates the relative meaning of RMSE. Thus, MAPE, and the visual depiction in Figure 7, are more useful to determine model performance. For both SARIMAX and VARX, we notice that the predictions are continuing in the opposite direction of the observed data trend. Therefore, XGBoost is more valuable as a predictor for TRANSIT. It closely follows the trend and direction of the test data (last two-years).

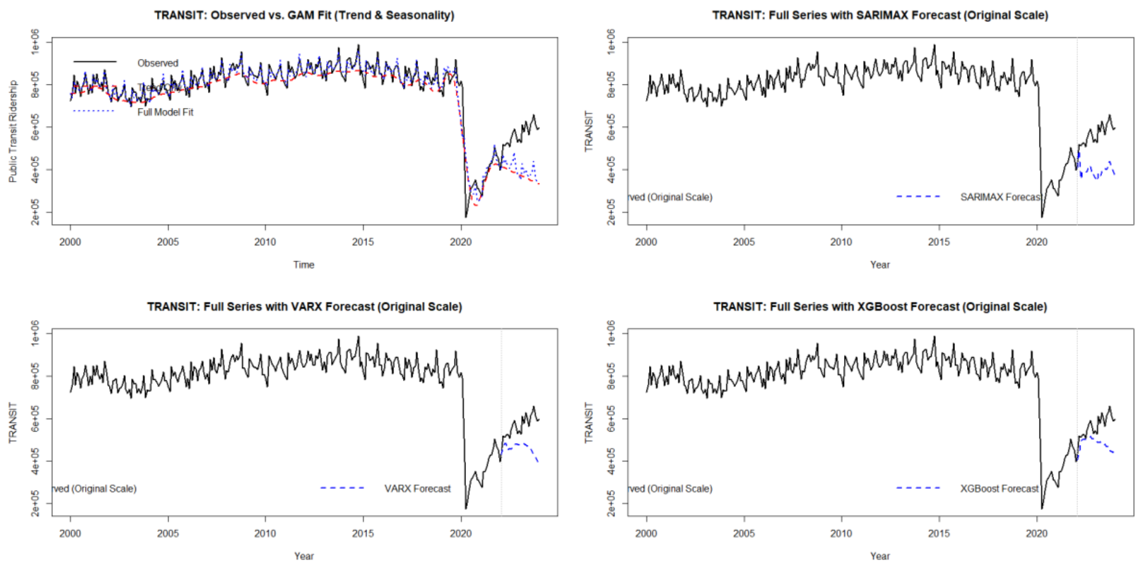

Figure 8: Comparison of model predictive performance for TRANSIT models (original scale)

Table 4 below summarizes the top features driving the performance of the XGBoost models.

Table 4: The ranking of the top three features identified by the XGBoost algorithm.

Response	#1 Feature	#2 Feature	#3 Feature
VMT	Lag12	Lag1	Month
AMVPNO	Lag1	Year	AISRSA
TRANSIT	Lag1	COVID	AUENSA

As many of these features were artificially created, a general interpretation is as follows. For VMT, the primary feature of Lag12 suggests the variable experiences high seasonality, as its value is most dependent on the results of a prior calendar year. This is consistent with the increased performance seen in SARIMAX. For AMVPNO, the primary feature of Lag1 indicates that the variable is strongly influenced by the previous month's value, indicating a high degree of autocorrelation. This is consistent with the fact that the auto.arima() determined seasonality did not improve the model's performance and retained a moving average order of 3. Similarly, TRANSIT is also highly dependent on the previous month. Unlike other response variables, the other observed timeseries, such as the COVID variable and AUENSA, played a higher degree of importance in whether public transportation was utilized.

Supplemental Analysis to Framework 1 - SARIMAX-GARCH

Alongside Framework 1, we developed an independent supplementary forecasting module using a SARIMAX–GARCH procedure. This aids us to better understand the statistical behaviour of VMT and its sensitivity to macro-economic variables. We assessed multicollinearity among the candidate explanatory variables using the Variance Inflation Factor (VIF). This step allowed us to confirm the stability of the exogenous variables (AISRSA, AUENSA, GASREGCOVM, TRANSIT, TOTALSA, NATURALGASD11, VIF<=10) and to avoid inflated coefficient uncertainty arising from redundant predictors. After aligning VMT with the selected regressors, we estimated a SARIMAX model using automatic ARIMA order selection. The Ljung-Box test for serial correlation and ARCH testing for conditional variance clustering detected non-white noise residuals, similar to the issues encountered with TRANSIT modeling. To correct, we fit a GARCH(1,1) model to the ARIMA residuals, then proceeded to forecast a 24-month period. The results showed that the model was able to reproduce the recent trajectory of VMT within stable predictive bounds (Figure 9). To evaluate resistance to exogenous shocks, we introduced up to a 20% upward shock on exogenous factors and generated 400 simulated SARIMAX–GARCH paths to quantify the expected shift in VMT. Although exploratory, this analysis provides directional insight into how VMT might respond under alternative macro-economic conditions.

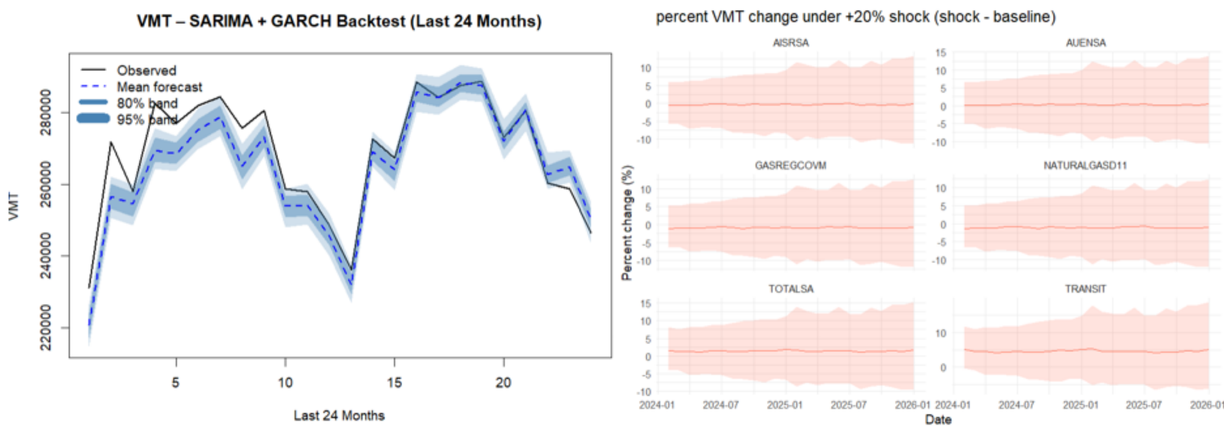

Figure 9: SARIMA-GARCH BackTest for last 24 months (left) and the expected shift in VMT and the associated uncertainty for each exogenous variable (right).

<u>Framework 2</u>

The XGBoost model developed for forecasting TOTALSA demonstrated strong predictive performance and proved to be effective at capturing the non-linear structure and the complex lag dynamics that were present in the time series. After training the model using lagged values, calendar based features (year and month), and the given exogenous variables, using the optimum obtained parameters shown in Table 5 (in Graphics), the model was evaluated on a two year test set. The test Root Mean Squared Error (RMSE) was 0.5403, indicating that the model's average deviation from true monthly sales remained relatively small. RMSE penalizes larger mistakes more heavily, so a value near 0.54 suggests the model rarely produced large forecasting errors. The Mean Absolute Error (MAE) of 0.4208 shows that most predictions stayed within roughly half a unit of the true outcome. In percentage terms, the model's Mean Absolute Percentage Error (MAPE) was 0.0267, meaning the XGBoost forecasts were off by only 2.67% on average, which is an exceptionally strong performance for an economic time series characterized by cyclical behavior, structural breaks, and pandemic-era volatility. This low MAPE indicates that the model provided highly accurate directional and magnitude predictions even with volatile periods.

We can see in Figure 10 below that the model tracked the general shape and direction of the test period with minimal lag. Unlike linear or autoregressive models, XGBoost displayed the ability to adjust to abrupt changes in market conditions, highlighting the benefit of non-linear modeling when forecasting sales influenced by complex economic dynamics.

Figure 10: XGBoost forecast for TOTALSA versus actual values.

The plot above together with the evaluation metrics indicates that XGBoost effectively captured both seasonal structure and nonlinear relationships among predictors by maintaining an accurate forecast across the two year test window despite periods of structural disruption such as the COVID-19 recovery period. This affirms the suitability of using an XGBoost algorithm for modeling complex economic dynamics in vehicle sales.

The XGBoost feature importance results for the TOTALSA forecasting model can be seen in Figure 11 below and in Table 6 (in Appendix). The results reveal a heavily autoregressive structure, with past values of TOTALSA overwhelmingly driving predictive performance. The most influential predictor was the 1-month lag of total sales, which accounted for 77.7% of total model gain, meaning that this feature was the main contributor in the model's total reduction in forecasting error. This tells us that the most recent month's sales remain the dominant signal in explaining current sales levels. The second most important feature was the total sales year-over-year percentage change, which contributed 6.8% of total gain and had the highest cover value of the top predictors (0.172) suggesting that while it was not as impactful as the one-month lag, it informed a broad subset of decision-tree splits and helped the model generalize across the full range of observations. Additional lagged features of total sales including lag 2, lag 3, lag 6, and lag 12 together contributed another meaningful share of importance. The presence of lag 12, with a gain of 2.16%, indicates that the model captured annual seasonality patterns in total vehicle sales.

Exogenous automotive and economic indicators such as AMVPNO (manufacturer's new orders), GASREGCOVM (conventional gas price), DAUPSA (domestic auto production in USA), and VMT (vehicle miles traveled) appeared in the model with small but nontrivial importance values. For example, AMVPNO lag 6 and lag 1 contributed 0.66% and 0.59% of gain, respectively, demonstrating that broader supply chain signals play some role in shaping sales expectations, however this was far less than internal dynamics of the sales series itself. Gasoline price lag 1 was also amongst the top 15 predictors with a 0.56% gain, suggesting that fuel costs impart modest predictive influence on consumer vehicle purchases. Measures of production, exports, transit usage, and natural gas consumption contributed small but diverse amounts of gain, collectively reinforcing that macroeconomic and transportation-related variables matter, but only marginally when compared with the dominant influence of past sales values. The importance of the exogenous variables and their lags is visualized in Figure 15 (in Graphics).

Calendar based variables like months and quarters held extremely low gain values (each below 0.01%), confirming that seasonality was captured through lagged values rather than explicit calendar variables. Their very low frequency values also indicate that the model rarely split on these variables, likely because the more informative lags already encoded seasonal structure. Together, the distribution of gain, cover, and frequency values suggests that XGBoost developed a forecasting strategy centered almost entirely on short-term autocorrelation, yearly seasonal recurrence, and minor adjustments from industry wide economic indicators, which is consistent with the strong predictive performance observed in the error metrics.

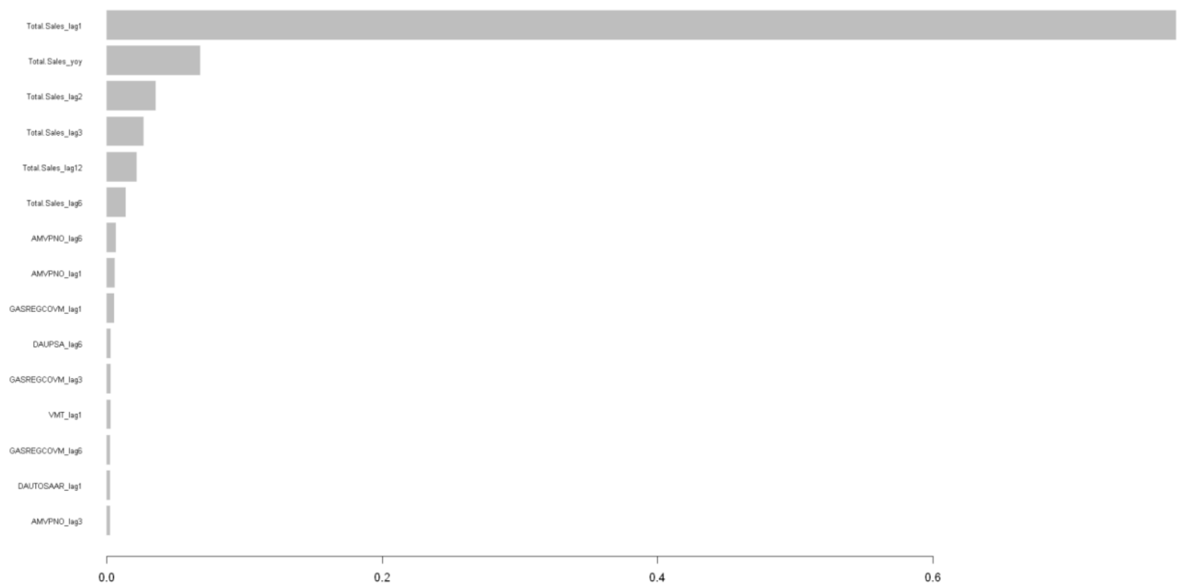

Figure 11: Comparison of Feature importance gains for XGBoost in Framework 2.

Validation

<u>Framework 1</u>

Performance of these models were validated through a two-year test data validation. Two-years were purposefully selected to incorporate the extreme effects seen from the COVID-19 pandemic. Since the United States government declared a national emergency from March 2020 to May 2023. As the datasets only include observations up until 2024, the test set includes the last year of the COVID-19 pandemic (2023), as well as the first year of economic reconstruction (2024). The test data's RMSE and MAPE, reported in Table 3, confirm the superior predictive power of the machine learning algorithm. XGBoost consistently exhibited the lowest error (VMT MAPE: 1.45%) due to its ability to support high-dimensional features. The performance of SARIMAX and VARX models was strong, with MAPE generally staying below 3%. This is a considerably high level of performance for economic forecasting, especially for a two-year prediction period.

<u>Framework 2</u>

Model validation was conducted using a 2-year test set to ensure that the forecast reflected true out-of-sample generalization. Metrics such as RMSE and MAPE were used to assess predictive accuracy, while graphical diagnostics evaluated directional correctness and responsiveness to turning points. The relatively low MAPE (2.67%) suggests strong performance for a macroeconomic forecasting problem. XGBoost's ability to model shifts from pandemic recovery to stabilized market conditions confirms the benefit of flexible, non-linear modeling for Total Sales. Additionally, cross-validation during hyperparameter tuning ensured that the selected configuration generalized well across different temporal folds.

Graphics

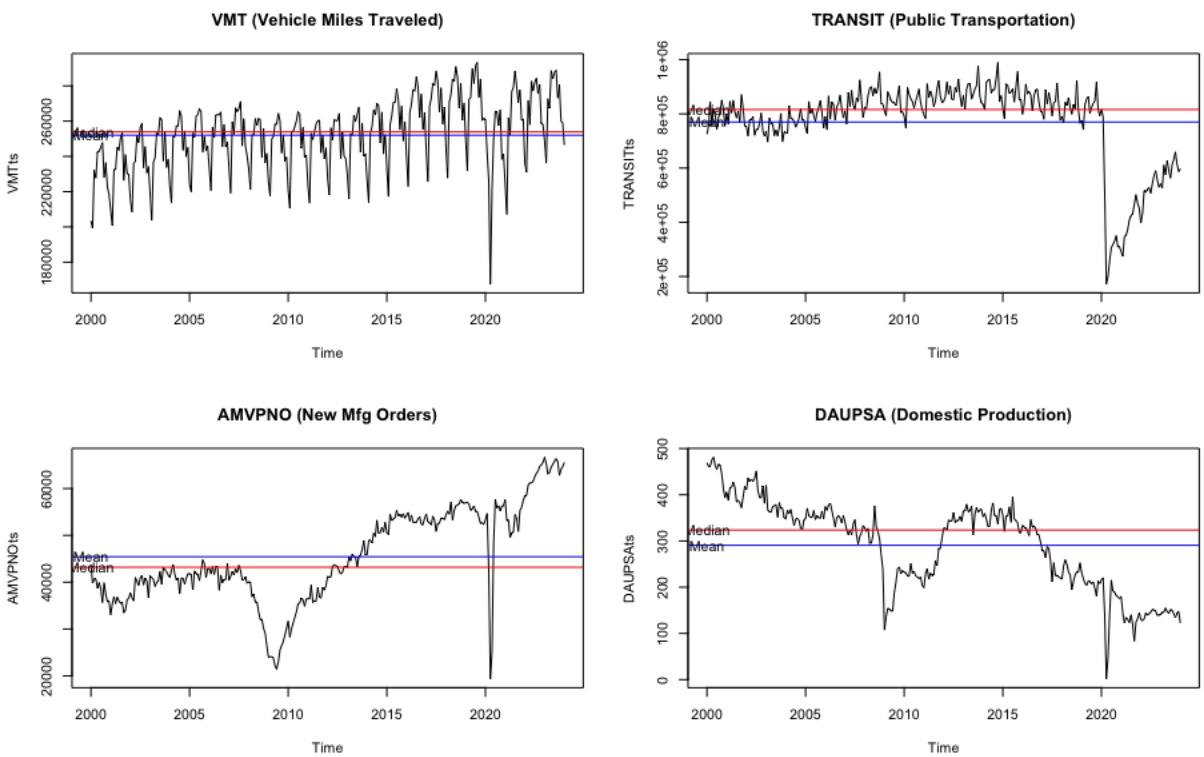

Figure 12: A time series plot showing the median (red) and mean (blue) of all variables. (Part 1)

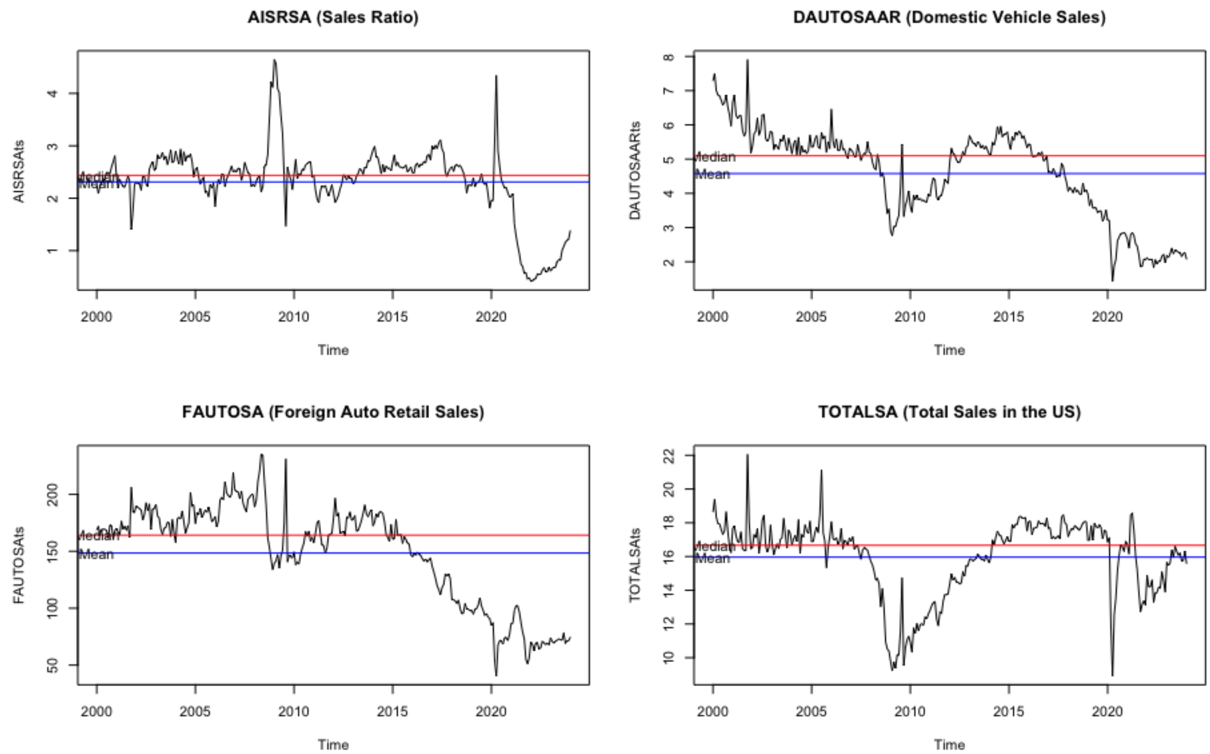

Figure 13: A time series plot showing the median (red) and mean (blue) of all variables. (Part 2)

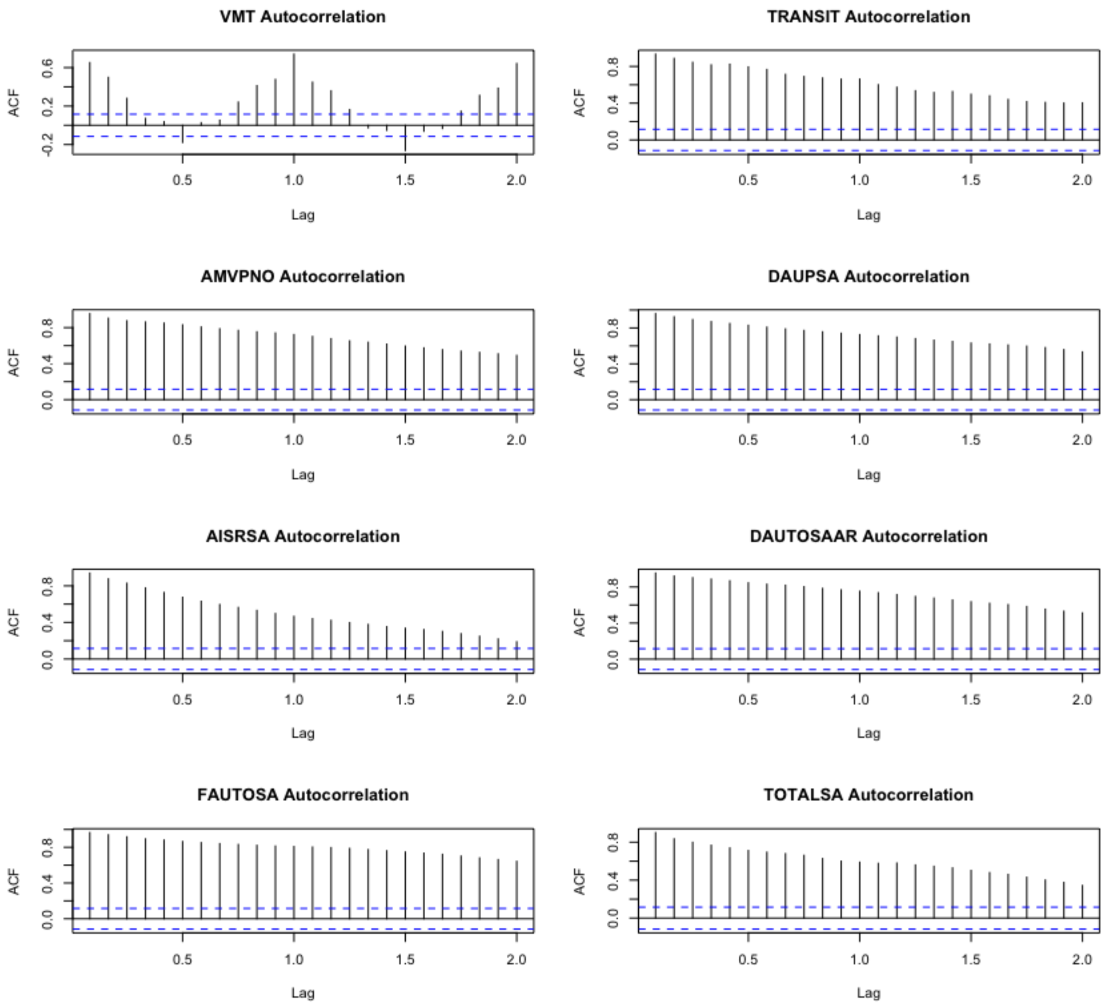

Figure 14: ACF plots of all examined variables, pre-treatment.

Table 5: Best Parameters found for the XGBoost model in Framework 2

Booster	*Eta*	*Max Depth*	*Subsample*	*ColSample By Tree*	*Nrounds*
Gbtree	0.05	3	0.9	0.9	546

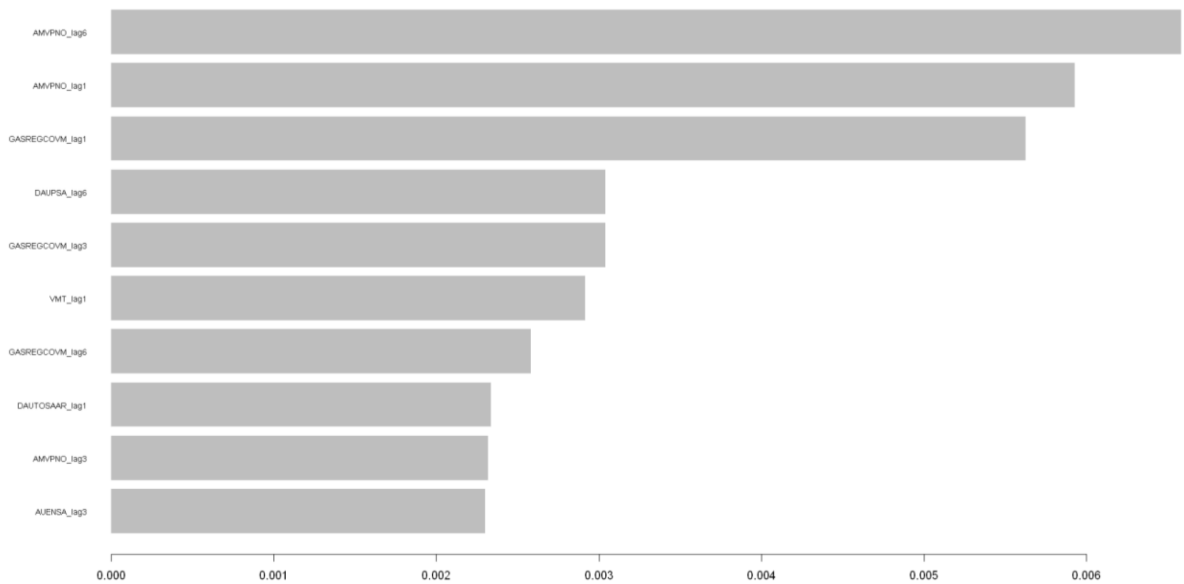

Figure 15: Comparison of Feature importance gains excluding Total Sales variables for XGBoost in Framework 2.

Explanation of Changes

Unlike our initial analysis plan which immediately planned to use SARIMAX and XGBoost, we opted to include both GAM and VARX. The purpose of using a GAM baseline was two-fold. First, it provided a performance benchmark. The high errors generated by GAM provided the justification for complex models like SARIMAX and VARX. Second, GAM provides a visual separation of trend, periodicity, and white noise. This gave an initial analysis step to identify if the COVID-19 pandemic, or other exogenous factors, were necessary to accurately model the response variable.

The purpose of including VARX was to determine if using a vectorized approach provided more flexibility in understanding. Initial analysis determined that seasonality may not be essential for all response variable modeling, so VARX provided another mechanism to model these variables.

Additionally, for the second framework analysis feature engineering was expanded beyond the original plan to include both autoregressive lags and lagged exogenous variables. An extensive hyperparameter search was also implemented to optimize model performance. These changes reflect the need for a robust machine-learning approach when modeling a highly volatile economic indicator like vehicle sales.

Conclusions

This study applied four different methodologies to model the impact of vehicle supply and exogenous factors to usage: GAM, SARIMAX, VARX, and XGBoost. The analysis consistently demonstrated that while forecasting performance increased, the ability to interpret relationships between variables decreased due to the increased complexity.

An exploration of VMT, a proxy for US vehicle utilization, consistently demonstrated a negative connection to public transportation usage (TRANSIT). During the pre-pandemic era, an increase in public transport logically correlated with a decrease in vehicle usage. In high density cities with large public transport networks, such as New York City, we would expect all models to faithfully predict the public's vehicle usage.

Car manufacturer orders (AMVPNO) were shown to be faithfully modeled by the automotive inventory-to-sales ratio (AISRSA). Manufacturers tend to increase vehicle orders when the inventory is low to support future sales. This confirms the basic economic feedback loop of supply and demand gradually influencing economic decisions in both directions.

Public transportation usage is the variable most susceptible to exogenous variables. The COVID-19 pandemic directed a substantial decrease in public transportation due to governmental policy, showing that governmental mandates can greatly impact both public activity and economic activity.

XGBoost effectively captured the short-term dynamics of US vehicle sales, demonstrating how production activity, fuel prices, and public transit usage influence purchasing behavior when encoded through lagged and seasonal features. The model's success emphasizes the nonlinear and interaction driven nature of automotive sales, especially in a market affected by supply chain constraints, employment fluctuations, and fuel price volatility. The findings reinforce well established automotive economics that sales depend both on immediate production activity and annual purchasing cycles. XGBoost's superior forecasting performance suggests that policy or industry analysis would benefit from incorporating non-linear machine learning models, especially when the goal is short-term forecasting rather than strict interpretability.

Success

This study reveals that while machine learning algorithms provide significantly improved predictive ability, they remove the interpretability that is essential for policy making. Use of SARIMAX and VARX can guide how other exogenous variables, such as GASREGCOVM or AISRSA affect the target variables in direct and lagged relationships.

The higher performance of XGBoost suggests that use of non-linear methods would not only provide better estimates, but also interpretable results. Additional non-linear methods like Non-linear VAR would theoretically increase predictive performance and interpretability. Alternatively, the max difference between SARIMAX and XGBoost MAPE across all models was equivalently 1.4%; thus, at the expense of slight bias, these more traditional methods can be used and interpreted at the same time.

XGBoost, in particular, achieved a good performance for TOTALSA due to its ability to model nonlinear interactions between production, economic, and behavioral variables. Its tree ensembles capture sharp turning points that linear or ARIMA based models often miss, while lagged features provide a rich autoregressive structure without requiring differencing. The algorithm also handles high-dimensional feature sets effectively, which is especially important when incorporating multiple lagged exogenous inputs. This flexibility is reflected in its low MAPE of 2.67%, highlighting its suitability for modeling TOTALSA, where market behavior is highly nonlinear and influenced by complex, interacting factors.

Limitations

In terms of limitations, VARX consistently failed to yield an insignificant Ljung-Box p-value. Despite the additional transformations on TRANSIT, VARX failed to faithfully model the serial correlation between variables. Future work could use Vector Error Correction Model or Vectorized ARIMA to better model the error structure for these multivariate models.

Despite its high accuracy, the XGBoost model also has several limitations. Its limited interpretability means that feature importance alone does not reveal the direction or true marginal effect of predictors. The model is also sensitive to feature engineering, as performance depends heavily on the chosen lag structure. While lagged features help capture temporal patterns, the model does not explicitly handle autocorrelation in the way that SARIMAX or VARX do. Forecast accuracy also decreases over longer horizons due to error accumulation in lagged predictions. These limitations suggest that, although XGBoost is a powerful forecasting tool, it is best used in combination with interpretable models when evaluating policies or supply-chain scenarios.

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Appendix

Table 6: Feature Importance for Final XGBoost model in Framework 2

Feature	Gain	Cover	Frequency
Total.Sales_lag1	7.77E-01	5.45E-02	0.081364044
Total.Sales_yoy	6.80E-02	1.72E-01	0.125934789
Total.Sales_lag2	3.56E-02	2.67E-02	0.04157942
Total.Sales_lag3	2.69E-02	1.03E-02	0.024827999
Total.Sales_lag12	2.16E-02	1.18E-01	0.093030212
Total.Sales_lag6	1.36E-02	1.74E-02	0.030810649
AMVPNO_lag6	6.58E-03	1.34E-02	0.017349686
AMVPNO_lag1	5.93E-03	1.25E-02	0.018546216
GASREGCOVM_lag1	5.63E-03	2.19E-02	0.020939276
DAUPSA_lag6	3.04E-03	2.87E-02	0.025426264
GASREGCOVM_lag3	3.04E-03	3.28E-02	0.027819324
VMT_lag1	2.91E-03	3.79E-02	0.028118457
GASREGCOVM_lag6	2.58E-03	1.75E-02	0.019443614
DAUTOSAAR_lag1	2.34E-03	2.83E-02	0.023930601
AMVPNO_lag3	2.32E-03	1.44E-02	0.017648818
AUENSA_lag3	2.30E-03	1.79E-02	0.021537541
AISRSA_lag3	1.73E-03	2.29E-02	0.022135806
TRANSIT_lag1	1.67E-03	2.51E-02	0.027520191
AUENSA_lag6	1.66E-03	7.79E-03	0.012563566
TRANSIT_lag6	1.66E-03	2.05E-02	0.017050553

AISRSA_lag1	1.65E-03	2.10E-02	0.023930601
NATURALGASD11_lag3	1.33E-03	1.03E-02	0.009871373
AISRSA_lag6	1.26E-03	1.64E-02	0.017648818
DAUTOSAAR_lag6	1.22E-03	1.03E-02	0.015554891
AUENSA_lag1	1.22E-03	8.45E-03	0.016153156
VMT_lag6	9.06E-04	3.01E-02	0.027221059
FAUTOSA_lag6	8.94E-04	1.79E-02	0.017050553
FAUTOSA_lag1	8.90E-04	2.76E-02	0.025127131
FAUTOSA_lag3	7.68E-04	3.73E-02	0.027520191
DAUPSA_lag3	7.23E-04	1.00E-02	0.012264433
NATURALGASD11_lag1	6.87E-04	3.47E-02	0.027520191
DAUTOSAAR_lag3	4.49E-04	1.17E-02	0.010469638
NATURALGASD11_lag6	4.42E-04	1.43E-02	0.017050553
DAUPSA_lag1	3.82E-04	4.96E-03	0.010469638
TRANSIT_lag3	3.15E-04	1.49E-02	0.015854023
VMT_lag3	3.10E-04	1.13E-02	0.011965301
month6	1.23E-04	6.69E-03	0.004187855
month2	5.26E-05	7.52E-04	0.000897398
quarter	3.48E-05	5.76E-04	0.001794795
month11	2.58E-05	1.77E-04	0.000299133
month12	2.45E-05	3.92E-03	0.00239306
month1	2.32E-05	6.56E-04	0.00119653

month4	2.23E-05	5.89E-05	0.000299133
month10	1.80E-05	2.57E-03	0.001794795
month7	1.77E-05	1.50E-03	0.002093928
month5	9.55E-06	8.03E-04	0.000897398
month9	1.59E-06	6.69E-05	0.000299133
month3	7.68E-07	1.85E-04	0.000299133
month8	1.64E-07	1.61E-05	0.000299133